



BUSINESS PLAN

Digital Transformation of Primary Health Care (PHC) Centres in Enugu State

Project Title: PHC Digital Health System Pilot (5 Centres)
Proponent: Mr. Udensi Okechukwu

Scope: Design, development, deployment, and staff training



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1. Executive Summary

This proposal outlines the deployment of a **fully integrated digital health system** across five Primary Health Care (PHC) centres in Enugu State.

The solution will:

- Digitize patient records
- Improve drug inventory management
- Enable offline-first operation with periodic synchronization
- Support reporting compatible with national systems like DHIS2

A pilot across five centres will demonstrate scalability across all PHCs in the state.

2. Objectives

- Replace paper-based records with a secure and efficient Electronic Medical Records (EMR) system to improve data accuracy and accessibility
- Reduce drug stock-outs, expiries, and wastage through real-time inventory tracking and automated alerts
- Improve patient turnaround time and service delivery efficiency through streamlined digital workflows
- Provide real-time, data-driven reporting and analytics to the Ministry for informed decision-making and policy planning

- **Enhance government revenue and cost savings** by minimizing leakages, improving billing transparency, and reducing operational inefficiencies across PHC centres
- **Improve patient outcomes and satisfaction** through faster diagnosis, better record continuity, reduced waiting times, and improved access to care
- **Ensure uninterrupted healthcare service delivery** through 24-hour power-backed digital infrastructure
- **Create direct and indirect employment opportunities**, including:
 - IT support personnel (minimum of 1 per centre = 5 jobs)
 - System administrators and data officers (estimated 3–5 roles centrally)
 - Installation and technical deployment teams during rollout (10–15 temporary jobs)
 - Ongoing maintenance and support services

Estimated total employment impact: 15–25 jobs (*direct and indirect*)

- Establish a scalable digital health framework that can be expanded across all PHC centres in Enugu State

Each PHC centre will be delivered as a fully self-sustaining digital unit with guaranteed 24-hour power availability, ensuring uninterrupted healthcare service delivery regardless of grid conditions.

3. Selected Pilot PHC Centres

Proposed high-impact centres:

1. **Abakpa Nike PHC (Enugu East)**
2. **Uwani PHC (Enugu South)**
3. **Ogui PHC (Enugu North)**
4. **Emene PHC (Enugu East)**
5. **Nsukka Urban PHC (Nsukka LGA)**

(Final selection subject to Ministry approval.)

4. Solution Overview

Core Modules

- Patient Registration & EMR
- Pharmacy & Inventory
- Laboratory (basic)
- Antenatal & Immunization tracking
- Reporting dashboard

Technology Stack

- Backend: PHP / Node.js
- Database: MySQL / PostgreSQL
- Offline sync engine
- Integration-ready with OpenMRS (optional hybrid)

5. Infrastructure Setup (Per Centre)

Hardware Requirements

Laptops (3 per centre):

- Intel Core i5 (11th Gen or higher)
- 8GB RAM (16GB recommended)
- 512GB SSD
- 14–15.6” display
- Battery backup 5–7 hours

Local Server (Central)

- Rack/Tower Server
- Intel Xeon / AMD EPYC
- 32GB RAM
- 4TB RAID Storage
- UPS backup

Networking

- Router + Managed Switch
- Structured LAN cabling
- Secure Wi-Fi access
- VPN for remote support

6. Implementation Plan

Phase	Activity	Duration
Phase 1	Planning & Customization	2 weeks
Phase 2	Development & Testing	4 weeks
Phase 3	Installation	2 weeks
Phase 4	Training	1 week

Phase	Activity	Duration
Phase 5	Pilot Run & Support	8 weeks

7. Training Plan

Each centre will receive:

- 2-day staff training
- Admin training for supervisors
- User manuals and support access

Good call—this is **critical in PHCs** and will actually strengthen your proposal. The key is to present the solar not as an extra, but as **core infrastructure for system reliability**.

Here's how to include it **professionally within your ₦20M per centre** (rebalanced, not just added on top):



8. COST IMPLICATION

Cost Per PHC Centre: ₦20,000,000

A. Hardware, Server & IT Infrastructure

Item	Qty	Unit Cost (₦)	Total (₦)
High-performance Laptops (Core i5, 16GB RAM, 512GB SSD)	5	800,000	4,000,000
Local Edge Server (Dedicated Mini Server System)	1	2,200,000	2,200,000
UPS (Server-grade backup)	2	300,000	600,000
Networking (Router, Switch, Structured Cabling)	1	700,000	700,000

Subtotal: ₦7,500,000

B. Solar Power System (24-Hour Uptime – Critical Infrastructure)

Item	Qty	Unit Cost (₦)	Total (₦)
Solar Panels (3–5kW system)	1	2,000,000	2,000,000
Lithium Battery Bank (10–15kWh storage)	1	2,200,000	2,200,000
Hybrid Inverter System	1	800,000	800,000
Installation & Mounting	1	500,000	500,000

Subtotal: ₦5,500,000

This ensures:

- 24/7 uptime for server & network
- Protection against power fluctuations
- Reduced long-term fuel/generator costs

C. Software Development & Integration

Item	Cost (₦)
Custom EMR System Development	3,000,000
Offline Sync Engine	800,000
Database & Security Setup	700,000
Reporting + DHIS2 Integration	700,000

Subtotal: ₦5,200,000

D. Deployment & Implementation

Item	Cost (₦)
Installation & Configuration	1,000,000
Data Setup / Migration	400,000
Field Engineering	600,000

Subtotal: ₦2,000,000

E. Training, Support & Contingency

Item	Cost (₦)
Staff Training	700,000
Documentation	200,000
12-Month Support & Maintenance	1,000,000

Item	Cost (₦)
Contingency	900,000

Subtotal: ₦2,800,000

TOTAL PER CENTRE

₦20,000,000

TOTAL FOR 5 CENTRES

₦100,000,000

9. Risk & Mitigation Plan

Risk	Impact	Mitigation
Poor internet connectivity	Data sync delays	Offline-first system
Staff resistance	Low adoption	Hands-on training
Power outages	Downtime	UPS + backup
Data loss	Critical	Automated backups
Budget constraints	Delay	Phased rollout

10. Sustainability Plan

- Local IT admin training
- Annual maintenance contract
- System scalability to all PHCs

11. Conclusion

This project provides a **cost-effective, scalable digital health solution** that aligns with state and national health priorities. A successful pilot will position Enugu State as a leader in PHC digital transformation.

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